2 hours on examples (12+ hours on EGMO not included)

Problem 1 (Z7E9BBA7)

Since BC is a chord, note that X is the reflection over the midpoint of BC. Therefore,

$$x = b + c; y = a + c; z = a + b$$

Note that the midpoints of AX, BY, CZ concur at $\frac{a+b+c}{2}$.

Problem 3 (16PUMACFA3)

Let the points be on the unit circle.

By homothety, F is just the midpoint of (D and the reflection of C over AB). Therefore,

$$f = \frac{d + a + b - ab\overline{c}}{2}$$

Therefore,

$$\angle BMF = \angle CBD$$

$\Leftrightarrow \frac{m-b}{m-f} \div \frac{b-c}{b-d}$ is real and positive.

Note that,

$$z := \frac{m-b}{m-f} \div \frac{b-c}{b-d}$$

$$= \frac{a-b}{a+b-\left(d+a+b-\frac{ab}{c}\right)} \div \frac{b-c}{b-d}$$

$$= \frac{ac-bc}{ab-cd} \div \frac{b-c}{b-d}$$

Note that,

$$\overline{z} = \frac{\frac{1}{ac} - \frac{1}{bc}}{\frac{1}{ab} - \frac{1}{cd}} \div \frac{\frac{1}{b} - \frac{1}{c}}{\frac{1}{b} - \frac{1}{d}}$$

$$= \frac{bd-ad}{cd-ab} \div \frac{cd-bd}{cd-bc}$$

$$= \frac{a-b}{ab-cd} \cdot d \div \left(\frac{b-c}{b-d} \cdot \frac{d}{c}\right)$$

$$= \frac{ac-bc}{ab-cd} \div \frac{b-c}{b-d} = z$$

Therefore, we're done.

3 hours; 5 clubs

Problem 5 (China TST 2011)

We claim the two are oppositely similar.

Let (ABC) be the unit circle. We get,

$$a' = -a; d = \frac{1}{2}(a + b + p - bc\overline{p})$$

Therefore, $x = a + b + c + p - bc\overline{p}$. Therefore,

$$ABC \sim XYZ \text{ oppositely} \Leftrightarrow \frac{c-a}{b-a} = \frac{\overline{z-x}}{y-x}$$

$$\Leftrightarrow \frac{c-a}{b-a} = \frac{\overline{bc\overline{p} - ab\overline{p}}}{bc\overline{p} - ac\overline{p}}$$

Multiplying top and bottom of RHS by $\frac{1}{abc\cdot\overline{p}}$, we are done.

Problem 7 (Taiwan TST 2014)

We let the incircle be the unit circle. To avoid confusion, we overlap $O \coloneqq I$. We have $o = 0$.

We may also WLOG rotate to let $d = 1$. Using tangents on unit circle, we'd get,

$$b = \frac{2f \cdot 1}{f+1} = 2 - \frac{2}{f+1}; c = 2 - \frac{2}{e+1}$$

$$\therefore m = 2 - \frac{e+f+2}{(e+1)(f+1)}$$

On the other hand, using intersection of two chords formula with chord DD, we get,

$$q = \frac{hg(d+d) - d^2(h+g)}{hg - d^2}$$

$$= \frac{2hg - h - g}{hg - 1}$$

$$= 2 - \frac{h+g-2}{hg-1}$$

We apply the transformation $z \to 2 - z$, which preserves perpendicularity as it's a dilation followed by translation.

$$m' = \frac{e+f+2}{(e+1)(f+1)}; q' = \frac{h+g-2}{hg-1} = \frac{-e-f-2}{ef-1}; o' = 0$$

We want to prove, $O'Q'$ and $O'M'$ are perpendicular, so we prove,

$$\frac{m'}{q'} = -\overline{\frac{m'}{q'}}$$

$$\Leftrightarrow \frac{\frac{e+f+2}{(e+1)(f+1)}}{\frac{-e-f-2}{ef-1}} = -\overline{\frac{\frac{e+f+2}{(e+1)(f+1)}}{\frac{-e-f-2}{ef-1}}}$$

$$\Leftrightarrow -\frac{ef-1}{(e+1)(f+1)} = -\left(\overline{-\frac{ef-1}{(e+1)(f+1)}}\right)$$

$$\Leftrightarrow -\frac{ef-1}{(e+1)(f+1)} = \frac{\frac{1}{ef}-1}{\left(\frac{1}{e}+1\right)\left(\frac{1}{f}+1\right)}$$

This is obviously true.

Problem 9 (P17PAKTST1)

We let P' be the base of Q to AC.

We let a, b, c, d be points on the unit circle.

$$q = \frac{ad(b+c) - bc(a+d)}{ad - bc};$$

$$p = \frac{ac(b+d) - bd(a+c)}{ac - bd}$$

The problem additionally gives us that P is the base of the perpendicular from Q to AC. Through factoring, we get that,

$$p - \frac{a+b}{2} = -\frac{(a-b)(ac+bd-2cd)}{ac-bd}$$

The problem additionally gives us that P is the base of the perpendicular from Q to AC.

$$p = \frac{1}{2}(a + c + q - ac\overline{q})$$

$$= \frac{1}{2}\left(a + c + q - ac \cdot \frac{\frac{1}{ad}\left(\frac{1}{b}+\frac{1}{c}\right) - \frac{1}{bc}\left(\frac{1}{a}+\frac{1}{d}\right)}{\frac{1}{ad} - \frac{1}{bc}}\right)$$

$$= \frac{1}{2}\left(a + c + \frac{ad(b+c) - bc(a+d)}{ad-bc} - ac \cdot \frac{a+d-b-c}{ad-bc}\right)$$

$$= \frac{1}{2} \cdot \frac{a^2d - abc + acd - bc^2 + abd + acd - abc - bcd - a^2c - acd + abc + ac^2}{ad - bc}$$

$$= \frac{1}{2} \cdot \frac{a^2d + acd - bc^2 + abd - abc - bcd - a^2c + ac^2}{ad - bc}$$

$$= \frac{1}{2} \cdot \frac{a^2d + acd - bc^2 + abd - abc - bcd - a^2c + ac^2}{ad - bc}$$

$$\Leftrightarrow \frac{ac(b+d) - bd(a+c)}{ac - bd} = \frac{a^2d + acd - bc^2 + abd - abc - bcd - a^2c + ac^2}{2ad - 2bc}$$

$$\Leftrightarrow \frac{abc + acd - abd - bcd}{ac - bd} = \frac{a^2d + acd - bc^2 + abd - abc - bcd - a^2c + ac^2}{2ad - 2bc}$$

$$\Leftrightarrow 2a^2bcd + 2a^2cd^2 - 2a^2bd^2 - 2abcd^2 - 2ab^2c^2 - 2abc^2d + 2ab^2cd - 2b^2c^2d$$

$$= (a^3cd + a^2c^2d - abc^3 + a^2bcd - a^2bc^2 - abc^2d - a^3c^2 + a^2c^3) + (-a^2bd^2 - abcd^2 + b^2c^2d - ab^2d^2 + ab^2cd + b^2cd^2 + a^2bcd - abc^2d)$$

$$\Leftrightarrow ab^2cd - abcd^2 + 2a^2cd^2 - a^2bd^2 - 2ab^2c^2 + b^2c^2d = a^3cd + a^2c^2d - abc^3 - a^2bc^2 - a^3c^2 + a^2c^3 - ab^2d^2 + b^2cd^2$$

$$\Leftrightarrow (ab^2cd - a^3cd) + (a^2cd^2 - abcd^2) + (a^2cd^2 - b^2cd^2) + (ab^2d^2 - a^2bd^2) + (a^2bc^2 - ab^2c^2) + (a^3c^2 - ab^2c^2) + (b^2c^2d - a^2c^2d) + (abc^3 - a^2c^3) = 0$$

Since $a \neq b$, we factor out $a - b$ to get,

$$\Leftrightarrow -abcd - a^2cd + acd^2 + acd^2 + bcd^2 - abd^2 + abc^2 + a^2c^2 + abc^2 - ac^2d - bc^2d - ac^3 = 0$$

$$\Leftrightarrow (abc^2 - abcd) + (a^2c^2 - a^2cd) + (acd^2 - ac^2d) + (acd^2 - ac^3) + (bcd^2 - bc^2d) + (abc^2 - abd^2) = 0$$

Since $c \neq d$, we factor out $c - d$ to get,

$$\Leftrightarrow abc + a^2c - acd - ac^2 - acd - bcd + abc + abd = 0$$

$$\Leftrightarrow \mathbf{2abc - 2acd + a^2c - ac^2 - bcd + abd = 0}$$

Now note that,

$$PE \perp BC$$

$$\Leftrightarrow \frac{p - \frac{a+b}{2}}{b - c} + \overline{\frac{p - \frac{a+b}{2}}{b - c}} = 0$$

$$\Leftrightarrow -\frac{(a-b)(ac + bd - 2cd)}{(b-c)(ac-bd)} - \frac{cd(b-a)(bd + ac - 2ab)}{ad(c-b)(bd-ac)} = 0$$

Since $b \neq c; a \neq b; a \neq 0; bd - ac \neq 0$ as otherwise $BD \parallel AC$ so P doesn't exist.

$$\Leftrightarrow -(ac + bd - 2cd) + \frac{c(bd + ac - 2ab)}{a} = 0$$

$$\Leftrightarrow -a^2c - abd + 2acd - 2abc + bcd + ac^2 = 0$$

$$\Leftrightarrow PQ \perp AC$$

Therefore, we're done.

6 hours; 14 clubs

<u>Problem 10 (98SLG7)</u>

We let $d = 0; b = 1$. The problem gives, $c = -2; e = -a$. We are given,

$$\angle ACB = 2\angle ABC$$

$$\Leftrightarrow 2\angle ABC - \angle ACB = 0$$

$$\Rightarrow Im\left(\left(\frac{a-b}{c-b}\right)^2 \div \left(\frac{b-c}{a-c}\right)\right) = 0$$

$$\Leftrightarrow Im\left(\left(\frac{a-1}{-3}\right)^2 \div \left(\frac{3}{a+2}\right)\right) = 0$$

$$\Leftrightarrow Im\big((a-1)^2(a+2)\big) = 0$$

$$\Leftrightarrow Im(a^3 - 3a + 2) = 0$$

We start with proving,

$$\angle ECB \equiv 2\angle EBC \text{ mod } 180°$$

$$\Leftrightarrow Im\left(\left(\frac{e-c}{b-c}\right) \div \left(\frac{c-b}{e-b}\right)^2\right) = 0$$

$$\Leftrightarrow Im\left(\left(\frac{-a+2}{3}\right) \div \left(\frac{-3}{-a-1}\right)^2\right) = 0$$

$$\Leftrightarrow Im(a^3 - 3a - 2) = 0$$

$$\Leftrightarrow Im(a^3 - 3a + 2) = 0$$

This is the given. Clearly, it is impossible for, $2\angle EBC - \angle ECB \geq 360°$ since then $\angle EBC$ is too big. Similarly, it is impossible for $2\angle EBC - \angle ECB \leq -180°$ since then $\angle ECB$ is too big.

The only other case is if $\angle ECB = 2\angle EBC$. FTSOC, let this be possible. We'd have,

$$\angle ACE = \angle ACB + \angle ECB = 2\angle CBA + 2\angle EBC = 2\angle ABE$$

We reflect B over D to B'. Note that B' is the midpoint of CD. Since D is the midpoint of AC,

$$2\angle ABE = 2\angle AB'E = \angle ACE$$

This is clearly impossible as $\angle EAB' < \angle EAC$ and $\angle B'EA < \angle CEA$.

The only remaining possibility is $\angle ECB + 180° = 2\angle EBC$, so we are done.

Problem 12 (17MP4G19)

We let the sides be complex vectors that sum to 0.

$$s_1 + s_2 + \cdots + s_{13} = 0$$

WLOG, scale and rotate to let $s_1 = 1$. Since the exterior angles are also multiples of $20°$, we may get, for an increasing sequence $a_1, \dots, a_{12}$ where $1 \le a_1; a_{12} < 18$,

$$\Leftrightarrow \left(e^{\frac{i\pi}{18}}\right)^0 + \left(e^{\frac{i\pi}{18}}\right)^{a_1} + \left(e^{\frac{i\pi}{18}}\right)^{a_2} + \cdots + \left(e^{\frac{i\pi}{18}}\right)^{a_{12}} = 0$$

To take out the 1, we will need to use $a_i = 6; a_j = 12$ since 3 and 15 are positive and all others give irrationals that cannot sum to an integer. The remaining 10 terms must come in pairs that sum to zero for a similar reason. One construction is to have,

$$a_1 = 1; a_2 = 2; a_3 = 4; a_4 = 5; a_5 = 6; a_6 = 7; a_7 = 10; a_8 = 11; a_9 = 12; a_{10} = 13; a_{11} = 14; a_{12} = 16$$

Here,

$$\left(e^{\frac{i\pi}{18}}\right)^0 + \left(e^{\frac{i\pi}{18}}\right)^{a_5} + \left(e^{\frac{i\pi}{18}}\right)^{a_9} = 0;$$

$$\left(e^{\frac{i\pi}{18}}\right)^{a_1} + \left(e^{\frac{i\pi}{18}}\right)^{a_7} = 0;$$

And so on.

The corresponding angles are, in the following order,

$$160°, 160°, 140°, 160°, 160°, 160°, 120°, 160°, 160°, 160°, 160°, 140°, 140°$$

7.5 hours; 20 clubs

Problem 13 (19RMM2)

We may equivalently prove that, if P were the intersection of the tangents to Ω at D, E, then PA is tangent to Γ.

WLOG, we let (CDE) be the unit circle. We may also WLOG have DC and AB be perpendicular to the real axis. We get,

$$b = \overline{a}; a = 2e - c; d = \overline{c}$$

Using the tangents formula,

$$p = \frac{2}{\overline{e} + \overline{d}} = \frac{2e}{1 + ce}$$

We show that $\angle PAE = \angle ABE$ to prove PA tangency.

$$\Leftrightarrow \frac{a-p}{a-e} \div \frac{b-a}{b-e} \text{ is real}$$

$$\Leftrightarrow \frac{(2e-c)-\dfrac{2e}{1+ce}}{(2e-c)-e} \div \frac{\overline{a}-a}{\overline{a}-e} \text{ is real}$$

$$\Leftrightarrow \frac{\dfrac{(2e-c)+ce(2e-c)-2e}{1+ce}}{e-c} \div \frac{\overline{a}-a}{\dfrac{2}{e}-\dfrac{1}{c}-e} \text{ is real}$$

$$\Leftrightarrow \frac{(2ce^2-c^2e-c)\left(\frac{2}{e}-\frac{1}{c}-e\right)}{(e-c)(1+ce)(\overline{a}-a)} = \overline{\frac{(2ce^2-c^2e-c)\left(\frac{2}{e}-\frac{1}{c}-e\right)}{(e-c)(1+ce)(\overline{a}-a)}}$$

$$\Leftrightarrow \frac{(2ce^2-c^2e-c)\left(\frac{2}{e}-\frac{1}{c}-e\right)}{(e-c)(1+ce)(\overline{a}-a)} = \frac{c^2e^2\left(\frac{2}{ce^2}-\frac{1}{c^2e}-\frac{1}{c}\right)\left(2e-c-\frac{1}{e}\right)}{ce\left(\frac{1}{e}-\frac{1}{c}\right)ce\left(1+\frac{1}{ce}\right)(a-\overline{a})}$$

$$(2ce^2-c^2e-c)\left(\frac{2}{e}-\frac{1}{c}-e\right) = (2c-e-ce^2)\left(2e-c-\frac{1}{e}\right)$$

Multiplying both sides by ec to kill the fractions,

$$\Leftrightarrow (2ce^2-c^2e-c)(2c-e-ce^2) = (2c-e-ce^2)(2e^2c-c^2e-c)$$

We are hence done.

<u>Problem 15 (98SLG5)</u>

We let (ABC) be the unit circle.

$$h = a+b+c; d = b+c-bc\overline{a}; e = a+c-ac\overline{b}; f = a+b-ab\overline{c}$$

We apply the transformation $z \to h-z$ to each of the variables,

$$d' = a+\frac{bc}{a}; e' = b+\frac{ac}{b};\ f' = c+\frac{ab}{c}$$

DEF being collinear if and only if,

$$0 = \frac{i}{4}\begin{vmatrix} d' & \overline{d'} & 1 \\ e' & \overline{e'} & 1 \\ f' & \overline{f'} & 1 \end{vmatrix}$$

$$\Leftrightarrow 0 = \begin{vmatrix} a+\dfrac{bc}{a} & \dfrac{1}{a}+\dfrac{a}{bc} & 1 \\ b+\dfrac{ac}{b} & \dfrac{1}{b}+\dfrac{b}{ac} & 1 \\ c+\dfrac{ab}{c} & \dfrac{1}{c}+\dfrac{c}{ab} & 1 \end{vmatrix}$$

$$\Leftrightarrow \left(\frac{1}{a}+\frac{a}{bc}\right)\left(b+\frac{ac}{b}\right)+\left(\frac{1}{b}+\frac{b}{ac}\right)\left(c+\frac{ab}{c}\right)+\left(\frac{1}{c}+\frac{c}{ab}\right)\left(a+\frac{bc}{a}\right) = \left(a+\frac{bc}{a}\right)\left(\frac{1}{b}+\frac{b}{ac}\right)+\left(b+\frac{ac}{b}\right)\left(\frac{1}{c}+\frac{c}{ab}\right)+\left(c+\frac{ab}{c}\right)\left(\frac{1}{a}+\frac{a}{bc}\right)$$

$$\Leftrightarrow \sum_{cyc} \frac{b}{a} + \frac{c}{b} + \frac{a}{c} + \frac{a^2}{b^2} = \sum_{cyc} \frac{a}{b} + \frac{b}{c} + \frac{c}{a} + \frac{b^2}{a^2}$$

$$\Leftrightarrow 3\sum_{cyc} \frac{a}{b} - \frac{b}{a} = \sum_{cyc} \frac{a^2}{b^2} - \frac{b^2}{a^2}$$

$$\Leftrightarrow 3\frac{(a-b)(b-c)(c-a)}{abc} = \frac{(a^2-b^2)(b^2-c^2)(c^2-a^2)}{a^2b^2c^2}$$

Since $a, b, c \neq 0$ and $a \neq b \neq c \neq a$,

$$\Leftrightarrow 3 = \frac{(a+b)(b+c)(c+a)}{abc}$$

$$\Leftrightarrow 3 = 2 + \frac{a}{b} + \frac{b}{a} + \frac{b}{c} + \frac{c}{b} + \frac{c}{a} + \frac{a}{c}$$

$$\Leftrightarrow 4 = (a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$$

$$\Leftrightarrow |h|^2 = OH^2 = 4$$

This is as desired.

8.5 hours; 26 clubs

<u>Problem 16 (ZB7989B4 Brokard Theorem)</u>

Let a, b, c, d be points on the unit circle.

$$r = \frac{ac(b+d) - bd(a+c)}{ac - bd};$$

$$p = \frac{ab(c+d) - cd(a+b)}{ab - cd};$$

$$q = \frac{ad(b+c) - bc(a+d)}{ad - bc}$$

We can instead show $OR \perp PQ; OQ \perp RP; OP \perp RQ$. We start with,

$$OR \perp PQ$$

$$\Leftrightarrow Re\left(\frac{r}{p-q}\right) = 0$$

$$\Leftrightarrow Re\left(\frac{\frac{ac(b+d) - bd(a+c)}{ac - bd}}{\frac{ab(c+d) - cd(a+b)}{ab - cd} - \frac{ad(b+c) - bc(a+d)}{ad - bc}}\right) = 0$$

$$\Leftrightarrow Re\left(\frac{\frac{\left(ac(b+d)-bd(a+c)\right)(ab-cd)(ad-bc)}{ac-bd}}{\left(ab(c+d)-cd(a+b)\right)(ad-bc)-\left(ad(b+c)-bc(a+d)\right)(ab-cd)}\right)=0$$

$$\Leftrightarrow Re\left(\frac{\frac{(abc+acd-abd-bcd)(ab-cd)(ad-bc)}{ac-bd}}{(abc+abd-acd-bcd)(ad-bc)+(-abd-acd+abc+bcd)(ab-cd)}\right)=0$$

$$\Leftrightarrow Re\left(\frac{\frac{(abc+acd-abd-bcd)(ab-cd)(ad-bc)}{ac-bd}}{a^2bcd+a^2bd^2-a^2cd^2-abcd^2-ab^2c^2-ab^2cd+abc^2d+b^2c^2d-a^2b^2d-a^2bcd+a^2b^2c+ab^2cd+abcd^2+ac^2d^2-abc^2d-bc^2d^2}\right)=0$$

$$\Leftrightarrow Re\left(\frac{\frac{(abc+acd-abd-bcd)(ab-cd)(ad-bc)}{ac-bd}}{a^2bd^2-a^2cd^2-ab^2c^2+b^2c^2d-a^2b^2d+a^2b^2c+ac^2d^2-bc^2d^2}\right)=0$$

$$\Leftrightarrow Re\left(\frac{\frac{(abc+acd-abd-bcd)(ab-cd)(ad-bc)}{ac-bd}}{(a^2bd^2-a^2b^2d)+(a^2b^2c-a^2cd^2)+(ac^2d^2-ab^2c^2)+(b^2c^2d-bc^2d^2)}\right)=0$$

$$\Leftrightarrow Re\left(\frac{\frac{(abc+acd-abd-bcd)(ab-cd)(ad-bc)}{ac-bd}}{(b-d)(bc^2d-a^2bd+a^2bc-abc^2+a^2cd-ac^2d)}\right)$$

$$\Leftrightarrow \frac{(ab-cd)(ad-bc)}{(ac-bd)(b-d)(a-c)}+\overline{\frac{(ab-cd)(ad-bc)}{(ac-bd)(b-d)(a-c)}}=0$$

$$\Leftrightarrow \frac{(ab-cd)(ad-bc)}{(ac-bd)(b-d)(a-c)}+\frac{abcd\left(\frac{1}{ab}-\frac{1}{cd}\right)abcd\left(\frac{1}{ad}-\frac{1}{bc}\right)}{abcd\left(\frac{1}{ac}-\frac{1}{bd}\right)bd\left(\frac{1}{b}-\frac{1}{d}\right)ac\left(\frac{1}{a}-\frac{1}{c}\right)}=0$$

$$\Leftrightarrow \frac{(ab-cd)(ad-bc)}{(ac-bd)(b-d)(a-c)}+\frac{(cd-ab)(bc-ab)}{(bd-ac)(d-b)(c-a)}=0$$

This is obviously true. Since these steps never required $ABCD$ to be convex, we can swap variables around WLOG to get the other two conditions.

Problem 17 (11MEX2)

We let Y and X be the projections of B onto AC and ℓ respectively. Notice that Y is the midpoint of AE and X is the midpoint of AD.

We let a, b, c be points on the unit circle. We get,

$$x=\frac{1}{2}\left(a+a+b-a^2\overline{b}\right);y=\frac{1}{2}\left(a+b+c-ac\overline{b}\right)$$

$$\Rightarrow d=a+b-\frac{a^2}{b};e=b+c-\frac{ac}{b}$$

We prove that $h = a + b + c, e, d$ are collinear. We first apply the translation,

$$z \to h - z$$

We get,

$$h' = 0;$$

$$d' = \frac{a^2}{b} + c;$$

$$e' = a + \frac{ac}{b}$$

Therefore, they are collinear if and only if,

$$\frac{d'}{e'} = \overline{\frac{d'}{e'}}$$

$$\Leftrightarrow \frac{\frac{a^2}{b} + c}{a + \frac{ac}{b}} = \overline{\frac{\frac{a^2}{b} + c}{a + \frac{ac}{b}}}$$

$$\Leftrightarrow \frac{a^2 + bc}{ab + ac} = \frac{\frac{1}{a^2} + \frac{1}{bc}}{\frac{1}{ab} + \frac{1}{ac}}$$

$$\Leftrightarrow \frac{a^2 + bc}{b + c} = \frac{\frac{1}{a} + \frac{a}{bc}}{\frac{1}{ab} + \frac{1}{ac}} = \frac{bc + a^2}{c + b}$$

We are done.

10 hours; 34 clubs